Venkata Lakshmi Parimala Pasupuleti

Partner Platform Engineer · Distributed Systems · API Design & Contracts · Multi-Provider LLM Integration · Python · FastAPI · Production Operations

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SUMMARY

Senior Software Engineer with 8+ years building and operating enterprise API/integration systems that other teams depend on. At Entrata (100K+ users), owned multi-tenant SaaS APIs, third-party integrations, and async distributed systems in a regulated production environment where API contract correctness directly affected tenant eligibility and financial calculations. Built **AgentProbe**, an open-source evaluation toolkit routing across five LLM providers - OpenAI, Anthropic/Claude, Azure OpenAI, Groq, and Ollama - the exact multi-provider model lab ecosystem Baseten connects. Deployed an **MCP server** at Magizhs connecting agentic pipeline outputs to downstream enterprise tooling: authentication, execution, failure routing, and observability across the integration boundary. Built **Leasel**, an 8-agent LangGraph production system with trace-level observability, CI regression harness, and LLM-as-judge scoring. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Distributed Systems | API Design & Contracts | Partner Integrations | Multi-Provider LLM Routing | MCP / Agent Connectors | Model Serving Infrastructure | Python · FastAPI | Authentication & Identity | Observability | Production Operations | AWS · Azure · GCP | Docker · Kubernetes | CI/CD · GitHub Actions | LangGraph · LangChain | REST APIs · OpenAPI | Java · TypeScript · PHP | PostgreSQL · Redis | RabbitMQ · Kafka

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data analysis pipelines, and API integration layers connecting AI components to backend and frontend systems.
- Designing application-facing APIs that surface insights from raw business data into reporting layers with defined SLOs and failure handling.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- **Deployed an MCP server** connecting agentic pipeline outputs to downstream enterprise tooling - handled authentication, execution, failure routing, and observability across the integration boundary; defined API contracts and retry semantics for cross-system reliability.
- Built an agentic mission-planning system (LangGraph StateGraph: parse → plan → retrieve → validate → conditional routing) with **validation guardrails** enforcing hard constraints before export - non-compliant routes surface grounded rejection reasons to operators.
- Implemented a RAG layer over FAA Part 107 regulations with cited-source answers; engineered the full ingestion, chunking, and retrieval pipeline from source corpus to FastAPI endpoint.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ users, multi-tenant)

- **Owned third-party integrations and API contracts** across a multi-tenant SaaS platform serving 100K+ users - defined service boundaries, maintained API versioning, and resolved cross-team dependencies on integration correctness; zero compliance violations across 7 years.
- **Built and operated a RabbitMQ-based distributed async processing system** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees that eliminated double-processing; owned through production incidents and annual rent-schedule changes.
- **Led the Affordable Housing module's integration into the AWS S3 document migration** - inventoried document types, mapped legacy records to the framework's input contracts, validated metadata correctness post-cutover; 100% PHI compliance via Libsodium-encrypted pipelines, zero security incidents.
- **Owned the annual HUD income-limit import pipeline for 7 years** - few thousand federal rows → ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations; designed baseline validation, reconciliation, and rollback preparation for every release.
- Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring across multi-tenant production schemas; shipped 20+ full-stack modules across 120+ Agile sprints.
- Mentored 5+ engineers; led microservices redesign (Docker) achieving 50% scalability improvement and 30% deployment time reduction.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management modules (PHP, Python, Java, PostgreSQL) including income-limits import (1.6M HUD/Tax Credit records), document management, and compliance workflows.
- Led cloud migration of client documents from legacy storage to AWS S3; redesigned document storage architecture migrating hundreds of thousands of records to a modern schema.

PROJECTS

AgentProbe - Open-Source AI Evaluation Toolkit

2025 | github.com/venkatalakshmiparimala/AgentProbe

- **Routes across five LLM providers - OpenAI, Anthropic/Claude, Azure OpenAI, Groq, and Ollama** - the exact model lab/provider ecosystem Baseten's partner platform connects; dynamic routing with provider-level authentication and failure handling.
- 15-endpoint FastAPI backend; 5-metric eval engine (factual accuracy, hallucination rate, retrieval quality, completeness, compliance); CI-integrated regression harness with golden datasets and offline + online evaluation loops; zero vendor lock-in, runs fully locally.

Leasel - Agentic AI Support Investigation Platform

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD

- **8-agent LangGraph production system** with durable state, conditional routing, and HITL escalation - trace-level observability, LLM-as-judge scoring, and a CI regression harness catching degradations before they reach users.
- Enforced hard API boundaries in code: LLM reasons, deterministic logic or human makes every final regulated call; 93.6% precision and 87.9% F1 across 1,240 compliance test cases (10 rules, 5 US states + federal).

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017